

Supplementary Material for On the Convergence of Cyclic Hierarchical Federated Learning with Heterogeneous Data

APPENDIX ALGORITHMS

Cyclic Federated Learning(CFL) [1] divides C clients into M non-overlapping client groups. The groups and the order in which they are traversed by the central server are pre-determined and fixed throughout training to simulate a cyclic structure of client participation. In each global round, once one group becomes available, the server would select all or partial clients to train. Once selected, this group can not participate again at least for the next $M - 1$ global rounds. In Fig.1, we have settings for CFL as $C = 6, M = 3, N = 2$.

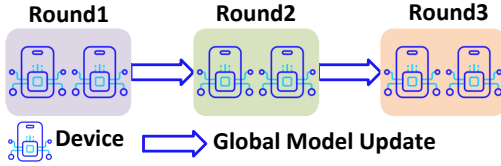


Fig. 1: Cyclic Federated Learning

Sequential Federated Learning(SFL) [2] sample clients without replacement randomly as the clients' training order. Within a round, each client initializes its model from its previous client and performs K local steps over its local dataset. Then, it passes its own model to the next client until all clients finish their local training. In Fig.2, we have process for SFL in one global round and set total clients number as $C = 6$.

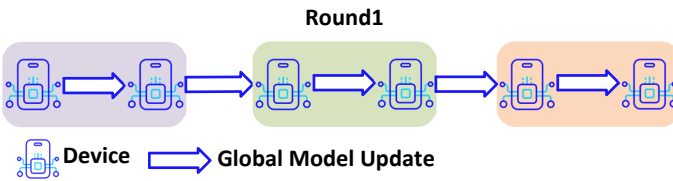


Fig. 2: Sequential Federated Learning

MODELS

The model architecture of the CNN used in the experiments is as follows.

TABLE I: CNN Architecture for MNIST

Layer Type	Size
Convolution+ReLu	5x5x10
Max Pooling	2x2
Convolution+ReLu	5x5x20
Max Pooling	2x2
Fully Connected+ReLU	320x50
Fully Connected	50x10

PROOFS

Proofs of Theorem 1 for Strongly Convex Objective

In this section, we give the proofs in detail for full and partial edge/client participation with HFL with cyclic pattern for three cases: strongly convex, general convex and non-convex. We will show them respectively. Our proofs are based on some identities and inequalities from C.2 and C.3 in [2], please check it for reference.

Lemma 1 *Let Assumptions 1, 2, 3 hold and assume that all the local objectives are μ -strongly convex. If the learning rate satisfies $\eta \leq \frac{1}{35LSPRTK}$, then it holds that*

$$\begin{aligned}
 & \mathbb{E} \left[\left\| \mathbf{x}^{(t+1)} - \mathbf{x}^* \right\|^2 \right] \\
 & \leq \left(1 - \frac{\mu SPKR\eta}{2} \right) \left\| \mathbf{x}^t - \mathbf{x}^* \right\|^2 \\
 & \quad + 7SPRK\eta^2\sigma^2 + 7S^2P^2R^2K^2\eta^2 \frac{(N-P)}{P(N-1)}\sigma_c^2 \\
 & \quad + \frac{21}{S(M-1)}\eta^2\sigma_g^2 - \frac{8}{5}SPKR\eta D_F(\mathbf{x}^t, \mathbf{x}^*) \\
 & \quad + 42LS^2P^2R^2K^2D_f(\mathbf{x}^t, \mathbf{x}^*) \\
 & \quad + (2L\eta + 7L^2SPRK\eta^2) \sum_{i,j,r,k} \mathbb{E} \left[\left\| \mathbf{x}_{r,k,t}^{i,j} - \mathbf{x}^t \right\|^2 \right]
 \end{aligned}$$

where $D_F(\mathbf{x}, \mathbf{x}^*) = \mathbb{E} \left[\left\| \nabla F_j(\mathbf{x}) - \nabla F_j(\mathbf{x}^*) \right\|^2 \right]$ and $D_f(\mathbf{x}, \mathbf{x}^*) = \mathbb{E} \left[\left\| \nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*) \right\|^2 \right]$.

Proof At t global round, the global model update of CHFL after one complete training round:

$$\Delta \mathbf{x} = \mathbf{x}^{t+1} - \mathbf{x}^t = -\eta \sum_{i=0}^{S-1} \sum_{j=0}^{P-1} \sum_{r=0}^{R-1} \sum_{k=0}^{K-1} \mathbf{g}_{r,k}^{i,j}$$

where $\mathbf{g}_{r,k}^{i,j} = \nabla F_j(\mathbf{x}_{r,k}^{i,j}, \xi_{r,k}^{i,j})$, thus:

$$\mathbb{E}[\Delta \mathbf{x}] = -\eta \sum_{i,j,r,k} \mathbb{E} \left[\nabla F_j(\mathbf{x}_{r,k}^{i,j}) \right]$$

We focus on a single training round and drop the superscripts t :

$$\begin{aligned}
 & \mathbb{E} \left[\left\| \mathbf{x} + \Delta \mathbf{x} - \mathbf{x}^* \right\|^2 \right] \\
 & = \left\| \mathbf{x} - \mathbf{x}^* \right\|^2 + 2\mathbb{E} \left[\langle \mathbf{x} - \mathbf{x}^*, \Delta \mathbf{x} \rangle \right] + \mathbb{E} \left[\left\| \Delta \mathbf{x} \right\|^2 \right] \\
 & = \left\| \mathbf{x} - \mathbf{x}^* \right\|^2 - 2\eta \underbrace{\sum_{i,j,r,k} \mathbb{E} \left[\left\langle \nabla F_j(\mathbf{x}_{r,k}^{i,j}), \mathbf{x} - \mathbf{x}^* \right\rangle \right]}_A
 \end{aligned}$$

$$+ \underbrace{\eta^2 \mathbb{E} \left[\left\| \sum_{i,j,r,k} g_{r,k}^{i,j} \right\|^2 \right]}_B$$

li2024Using Lemma 2 in [2] to bound A:

$$\begin{aligned} & -2\eta \sum_{i,j,r,k} \mathbb{E} \left[\left\langle \nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right), \mathbf{x} - \mathbf{x}^* \right\rangle \right] \\ & \leq -2\eta \sum_{i,j,r,k} \mathbb{E} [F_j(\mathbf{x}) - F_j(\mathbf{x}^*)] \\ & \quad + \frac{\mu}{4} \|\mathbf{x}^* - \mathbf{x}\|^2 - L \left\| \mathbf{x} - \mathbf{x}_{r,k}^{i,j} \right\|^2 \\ & \leq -2SPKR\eta D_F(\mathbf{x}, \mathbf{x}^*) - \frac{1}{2}\mu SPKR\eta \|\mathbf{x} - \mathbf{x}^*\|^2 \\ & \quad + 2L\eta \sum_{i,j,r,k} \left\| \mathbf{x} - \mathbf{x}_{r,k}^{i,j} \right\|^2 \end{aligned}$$

Bounding B using Jensen's inequality in C.2 [2], We observe that the three additional terms ⑤⑥⑦ are generated by the edge layer and the cyclic pattern.:

$$\begin{aligned} & \mathbb{E} \left[\left\| \sum_{i,j,r,k} g_{r,k}^{i,j} \right\|^2 \right] \\ & = \mathbb{E} \left[\left\| \sum_{i,j,r,k} \left\{ g_{r,k}^{i,j} - \nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) + \nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) - \nabla F_j(\mathbf{x}) \right. \right. \right. \\ & \quad \left. \left. + \nabla F_j(\mathbf{x}) - \nabla F_j(\mathbf{x}^*) + \nabla F_j(\mathbf{x}^*) + \nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*) \right. \right. \\ & \quad \left. \left. + \nabla f_i(\mathbf{x}^*) - \nabla f_i(\mathbf{x}) \right\} \right\|^2 \right] \\ & \leq \underbrace{7\mathbb{E} \left[\left\| \sum_{i,j,r,k} \left(g_{r,k}^{i,j} - \nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) \right) \right\|^2 \right]}_{\textcircled{1}} \\ & \quad + \underbrace{7\mathbb{E} \left[\left\| \sum_{i,j,r,k} \left(\nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) - \nabla F_j(\mathbf{x}) \right) \right\|^2 \right]}_{\textcircled{2}} \\ & \quad + \underbrace{7\mathbb{E} \left[\left\| \sum_{i,j,r,k} [\nabla F_j(\mathbf{x}) - \nabla F_j(\mathbf{x}^*)] \right\|^2 \right]}_{\textcircled{3}} \\ & \quad + \underbrace{7\mathbb{E} \left[\left\| \sum_{i,j,r,k} \nabla F_j(\mathbf{x}^*) \right\|^2 \right]}_{\textcircled{4}} \\ & \quad + \underbrace{7\mathbb{E} \left[\left\| \sum_{i,j,r,k} [\nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*)] \right\|^2 \right]}_{\textcircled{5}} \end{aligned}$$

$$+ \underbrace{7\mathbb{E} \left[\left\| \sum_{i,j,r,k} \nabla f_i(\mathbf{x}^*) \right\|^2 \right]}_{\textcircled{6}}$$

$$+ \underbrace{7\mathbb{E} \left[\left\| \sum_{i,j,r,k} (-\nabla f_i(\mathbf{x})) \right\|^2 \right]}_{\textcircled{7}}$$

Bounding ①:

$$7\mathbb{E} \left[\left\| \sum_{i,j,r,k} \left(g_{r,k}^{i,j} - \nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) \right) \right\|^2 \right] \leq 7SPRK\sigma^2$$

Bounding ②:

$$\begin{aligned} & 7\mathbb{E} \left[\left\| \sum_{i,j,r,k} \left(\nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) - \nabla F_j(\mathbf{x}) \right) \right\|^2 \right] \\ & \leq 7L^2SPRK \sum_{i,j,r,k} \mathbb{E} \left[\left\| \mathbf{x}_{r,k}^{i,j} - \mathbf{x} \right\|^2 \right] \end{aligned}$$

Bounding ③:

$$\begin{aligned} & 7\mathbb{E} \left[\left\| \sum_{i,j,r,k} [\nabla F_j(\mathbf{x}) - \nabla F_j(\mathbf{x}^*)] \right\|^2 \right] \\ & \leq 14LSPRK \sum_{i,j,r,k} \mathbb{E} [D_{F_j}(\mathbf{x}, \mathbf{x}^*)] \\ & \leq 14LS^2P^2R^2K^2D_F(\mathbf{x}, \mathbf{x}^*) \end{aligned}$$

Bounding ④:

$$\begin{aligned} & 7\mathbb{E} \left[\left\| \sum_{i,j,r,k} \nabla F_j(\mathbf{x}^*) \right\|^2 \right] \\ & \leq 7S^2R^2P^2K^2 \times \frac{N-P}{P(N-1)} \sigma_c^2 \end{aligned}$$

Bounding ⑤:

$$\begin{aligned} & 7\mathbb{E} \left[\left\| \sum_{i,j,r,k} [\nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*)] \right\|^2 \right] \\ & \leq 14LSPRK \sum_{i,j,r,k} \mathbb{E} [D_{f_i}(\mathbf{x}, \mathbf{x}^*)] \\ & \leq 14LS^2P^2R^2K^2D_f(\mathbf{x}, \mathbf{x}^*) \end{aligned}$$

Bounding ⑥:

$$\begin{aligned} & 7\mathbb{E} \left[\left\| \sum_{i,j,r,k} \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\ & \leq 7P^2K^2R^2\mathbb{E} \left[\left\| \sum_{i=0}^{S-1} \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\ & \leq 7P^2K^2R^2 \frac{(M-S)}{S(M-1)} \sigma_g^2 \end{aligned}$$

Bounding ①:

$$\begin{aligned}
& 7\mathbb{E} \left[\left\| \sum_{i,j,r,k} (-\nabla f_i(\mathbf{x})) \right\|^2 \right] \\
&= 7\mathbb{E} \left[\left\| \sum_{i,j,r,k} (-\nabla f_i(\mathbf{x}) + \nabla f_i(\mathbf{x}^*) - \nabla f_i(\mathbf{x}^*)) \right\|^2 \right] \\
&\leq 7\mathbb{E} \left[\left\| \sum_{i,j,r,k} (\nabla f_i(\mathbf{x}^*) - \nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*)) \right\|^2 \right] \\
&\leq 14\mathbb{E} \left[\left\| \sum_{i,j,r,k} (\nabla f_i(\mathbf{x}^*) - \nabla f_i(\mathbf{x})) \right\|^2 \right] \\
&\quad \underbrace{\hspace{10em}}_{\text{Based on ⑤}} \\
&\quad + 14\mathbb{E} \left[\left\| \sum_{i,j,r,k} \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\
&\quad \underbrace{\hspace{10em}}_{\text{Based on ⑥}} \\
&\leq 28LS^2P^2R^2K^2D_f(\mathbf{x}, \mathbf{x}^*) + \frac{14P^2K^2R^2(M-S)}{S(M-1)}\sigma_g^2
\end{aligned}$$

Then substituting above seven bounds into B:

$$\begin{aligned}
& \mathbb{E} \left[\left\| \sum_{i,j,r,k} g_{r,k}^{i,j} \right\|^2 \right] \\
&\leq 7SPRK\sigma^2 + 7L^2SPRK \sum_{i,j,r,k} \mathbb{E} \left[\left\| \mathbf{x}_{r,k}^{i,j} - \mathbf{x} \right\|^2 \right] \\
&\quad + 14LS^2P^2R^2K^2D_F(\mathbf{x}, \mathbf{x}^*) + 7S^2R^2 \times P^2K^2 \times \frac{N-P}{P(N-1)}\sigma_c^2 \\
&\quad + 14LS^2P^2R^2K^2D_f(\mathbf{x}, \mathbf{x}^*) + 7P^2K^2R^2 \frac{(M-S)}{S(M-1)}\sigma_g^2 \\
&\quad + 28LS^2P^2R^2K^2D_f(\mathbf{x}, \mathbf{x}^*) + \frac{14P^2K^2R^2(M-S)^2}{S(M-1)}\sigma_g^2 \\
&= 7SPRK\sigma^2 + 7L^2SPRK \sum_{i,j,r,k} \mathbb{E} \left[\left\| \mathbf{x}_{r,k}^{i,j} - \mathbf{x} \right\|^2 \right] \\
&\quad + 14LS^2P^2R^2K^2D_F(\mathbf{x}, \mathbf{x}^*) + 7S^2R^2P^2K^2 \times \frac{N-P}{P(N-1)}\sigma_c^2 \\
&\quad + 42LS^2P^2R^2K^2D_f(\mathbf{x}, \mathbf{x}^*) + \frac{21P^2K^2R^2S^2(M-S)}{S(M-1)}\sigma_g^2
\end{aligned}$$

Substituting A and B into following equation:

$$\begin{aligned}
& \mathbb{E} \left[\left\| \mathbf{x} + \Delta \mathbf{x} - \mathbf{x}^* \right\|^2 \right] \\
&\leq \left\| \mathbf{x} - \mathbf{x}^* \right\|^2 - 2\eta \sum_{i,j,r,k} \mathbb{E} \left[\left\langle \nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right), \mathbf{x} - \mathbf{x}^* \right\rangle \right] \\
&\quad + \eta^2 \mathbb{E} \left[\left\| \sum_{i,j,r,k} g_{r,k}^{i,j} \right\|^2 \right] \\
&\leq \left\| \mathbf{x} - \mathbf{x}^* \right\|^2 - 2SPKR\eta D_F(\mathbf{x}, \mathbf{x}^*)
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{2}\mu SPKR\eta \left\| \mathbf{x} - \mathbf{x}^* \right\|^2 + 2L\eta \sum_{i,j,r,k} \mathbb{E} \left[\left\| \mathbf{x} - \mathbf{x}_{r,k}^{i,j} \right\|^2 \right] \\
&+ 7SPRK\eta^2\sigma^2 + 7L^2SPRK\eta^2 \sum_{i,j,r,k} \mathbb{E} \left[\left\| \mathbf{x}_{r,k}^{i,j} - \mathbf{x} \right\|^2 \right] \\
&+ 14LS^2P^2R^2K^2D_F(\mathbf{x}, \mathbf{x}^*) \\
&+ 7S^2R^2 \times P^2K^2\eta^2 \times \frac{N-P}{P(N-1)}\sigma_c^2 \\
&+ 42LS^2P^2R^2K^2\eta^2D_f(\mathbf{x}, \mathbf{x}^*) \\
&+ \frac{21P^2K^2R^2S^2(M-S)}{S(M-1)}\eta^2\sigma_g^2 \\
&\leq \left(1 - \frac{\mu MPKR\eta}{2} \right) \left\| \mathbf{x} - \mathbf{x}^* \right\|^2 + 7SPRK\eta^2\sigma^2 \\
&+ 7S^2P^2R^2K^2\eta^2 \frac{(N-P)}{P(N-1)}\sigma_c^2 \\
&+ \frac{21}{S(M-1)}\eta^2\sigma_g^2 - \frac{8}{5}SPKR\eta D_F(\mathbf{x}, \mathbf{x}^*) \\
&+ 42LS^2P^2R^2K^2D_f(\mathbf{x}, \mathbf{x}^*) \\
&+ (2L\eta + 7L^2SPRK\eta^2) \underbrace{\sum_{i,j,r,k} \mathbb{E} \left[\left\| \mathbf{x}_{r,k}^{i,j} - \mathbf{x} \right\|^2 \right]}_{\text{client drift}}
\end{aligned}$$

Lemma 2 Let Assumptions 1,2,3 hold and assume that all the local objectives are μ -strongly convex. If the learning rate satisfies $\eta \leq \frac{1}{35LS^2P^2R^2K}$, then the client shift can be bounded as:

$$\begin{aligned}
\mathbb{E}_t \leq & \frac{71}{10}\eta^2q_B\sigma^2 + \frac{71}{5}Lq_{B^2}\eta^2D_F(\mathbf{x}, \mathbf{x}^*) + \frac{71}{10}q_c\eta^2\sigma_c^2 \\
& + 22q_g\eta^2\sigma_g^2 + 43Lq_{B^2}\eta^2D_f(\mathbf{x}, \mathbf{x}^*)
\end{aligned}$$

Proof Follow Lemma 6 in [2] to bound client drift:

$$\begin{aligned}
\mathbb{E}_t &= \sum_{i,j,r,k} \mathbb{E} \left[\left\| \mathbf{x}_{r,k}^{i,j} - \mathbf{x} \right\|^2 \right] \\
\text{where } r'(i) &= \begin{cases} R & i' < i-1 \\ r-1 & i' = i-1 \end{cases}, \quad j'(i) = \begin{cases} P & i' < i-1 \\ j & \text{the } j\text{th client.} \end{cases}, \quad k'(i) = \begin{cases} K-1 & i' < i-1 \\ k-1 & i' = i-1 \end{cases} \\
&\mathbb{E} \left[\left\| \mathbf{x}_{r,k}^{i,j} - \mathbf{x} \right\|^2 \right] \\
&\leq \mathbb{E} \left[\left\| -\eta \sum_{i',r',j',k'} g_{r',k'}^{i',j'} \right\|^2 \right] \\
&\leq \eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \left(g_{r',k'}^{i',j'} - \nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) + \nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) \right. \right. \right. \\
&\quad \left. \left. - \nabla F_j(\mathbf{x}) + \nabla F_j(\mathbf{x}) - \nabla F_j(\mathbf{x}^*) \right. \right. \\
&\quad \left. \left. + \nabla F_j(\mathbf{x}^*) + \nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*) + \nabla f_i(\mathbf{x}^*) - \nabla f_i(\mathbf{x}) \right\|^2 \right] \\
&\leq 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \left(g_{r',k'}^{i',j'} - \nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) \right) \right\|^2 \right]
\end{aligned}$$

$$\begin{aligned}
& + 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \left(\nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) - \nabla F_j(\mathbf{x}) \right) \right\|^2 \right] \\
& + 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} [\nabla F_j(\mathbf{x}) - \nabla F_j(\mathbf{x}^*)] \right\|^2 \right] \\
& + 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla F_j(\mathbf{x}^*) \right\|^2 \right] \\
& + 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} [\nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*)] \right\|^2 \right] \\
& + 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\
& + 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} (-\nabla f_i(\mathbf{x})) \right\|^2 \right] \\
& \leq 7\eta^2 \sum_{i',j',r',k'} \mathbb{E} \left[\left\| g_{r',k'}^{i',j'} - \nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) \right\|^2 \right] \\
& + 7\eta^2 B_{i,j,r,k} \sum_{i',j',r',k'} \mathbb{E} \left[\left\| \nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) - \nabla F_j(\mathbf{x}) \right\|^2 \right] \\
& + 7\eta^2 B_{i,j,r,k} \sum_{i',j',r',k'} \mathbb{E} \left[\left\| \nabla F_j(\mathbf{x}) - \nabla F_j(\mathbf{x}^*) \right\|^2 \right] \\
& + 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla F_j(\mathbf{x}^*) \right\|^2 \right] \\
& + 7\eta^2 B_{i,j,r,k} \sum_{i',j',r',k'} \mathbb{E} \left[\left\| \nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\
& + 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\
& + 14\eta^2 B_{i,j,r,k} \sum_{i',j',r',k'} \mathbb{E} \left[\left\| \nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\
& + 14\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\
& \leq 7B_{i,j,r,k}\eta^2\sigma^2 + 7L^2\eta^2 B_{i,j,r,k} \sum_{i',j',r',k'} \mathbb{E} \left[\left\| \mathbf{x}_{r',k'}^{i',j'} - \mathbf{x} \right\|^2 \right] \\
& + 14L\eta^2 B_{i,j,r,k}^2 D_F(\mathbf{x}, \mathbf{x}^*) \\
& + 7\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla F_j(\mathbf{x}^*) \right\|^2 \right] \\
& + 42L\eta^2 B_{i,j,r,k}^2 D_f(\mathbf{x}, \mathbf{x}^*)
\end{aligned}$$

$$+ 21\eta^2 \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla f_i(\mathbf{x}^*) \right\|^2 \right]$$

where $\sum_{i',j',r',k'} 1 = B_{i,j,r,k} = (i-1) RPK + (r-1) PK + k-1$ and $\sum_{i',j',r',k'} \sum_{i',j',r',k'} 1 = B_{i,j,r,k}^2$.

Then, returning to $\mathbb{E}_t = \sum_{i,j,r,k} \mathbb{E} \left[\left\| \mathbf{x}_{r,k}^{i,j} - \mathbf{x} \right\|^2 \right]$, we have

$$\begin{aligned}
\mathbb{E}_t & \leq 7\eta^2 \sigma^2 \sum_{i,j,r,k} B_{i,j,r,k} \\
& + 7L^2\eta^2 \sum_{i,j,r,k} B_{i,j,r,k} \sum_{i',j',r',k'} \mathbb{E} \left[\left\| \mathbf{x}_{r',k'}^{i',j'} - \mathbf{x} \right\|^2 \right] \\
& + 14L\eta^2 \sum_{i,j,r,k} B_{i,j,r,k}^2 D_F(\mathbf{x}, \mathbf{x}^*) \\
& + 7\eta^2 \underbrace{\sum_{i,j,r,k} \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla F_j(\mathbf{x}^*) \right\|^2 \right]}_{\textcircled{8}} \\
& + 42L\eta^2 \sum_{i,j,r,k} B_{i,j,r,k}^2 D_f(\mathbf{x}, \mathbf{x}^*) \\
& + 21\eta^2 \underbrace{\sum_{i,j,r,k} \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla f_i(\mathbf{x}^*) \right\|^2 \right]}_{\textcircled{9}}
\end{aligned}$$

As in [2], [3], the cyclic FL experiences client drift resulting from data heterogeneity among clients. If we only have client drift induced by cyclic pattern in FL, we only have one term $\textcircled{8}$ to be bounded, and there are only K and N that affect the gradient updates. In CHFL, we encounter additional edge model drift due to inter-edge heterogeneity. Then, we have an extra term $\textcircled{9}$ to be bounded. Moreover, for both $\textcircled{8}$ and $\textcircled{9}$, we have additional R and M , making the bounding formula more complex. Since we assume sampling without replacement for edges and clients, we follow Lemma 4 [2] with $\bar{\mathbf{x}} = \nabla F(\mathbf{x}^*) = 0$ for $\textcircled{8}$ and $\bar{\mathbf{x}} = \nabla f(\mathbf{x}^*) = 0$ for $\textcircled{9}$ to bound them.

$$\begin{aligned}
& \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*) \right\|^2 \right] \\
& = \mathbb{E} \left[\left\| \sum_{j=0}^{N-1} KR(i-1) [\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)] \right. \right. \\
& \quad + K(r-1) \sum_{j=0}^{N-1} [\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)] \\
& \quad \left. \left. + k [\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)] \right\|^2 \right] \\
& = K^2 R^2 (i-1)^2 \mathbb{E} \left[\left\| \sum_{j=0}^{N-1} [\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)] \right\|^2 \right]
\end{aligned}$$

$$\begin{aligned}
& + K^2(r-1)^2 \mathbb{E} \left[\left\| \sum_{j=0}^{N-1} (\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)) \right\|^2 \right] \\
& + k^2 \mathbb{E} \left[\|\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)\|^2 \right] \\
& + 2KR(i-1)K(r-1) \\
& \left[\left\langle \sum_{j=0}^{N-1} (\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)), \sum_{j=0}^{N-1} [\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)] \right\rangle \right] \\
& + 2KR(i-1)k\mathbb{E} \\
& \left[\left\langle \sum_{j=0}^{N-1} (\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)), (\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)) \right\rangle \right] \\
& + K(r-1)k\mathbb{E} \\
& \left[\left\langle \sum_{j=0}^{N-1} (\nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*)), \nabla F_j(\mathbf{x}^*) - \nabla F(\mathbf{x}^*) \right\rangle \right] \\
& \leq K^2 R^2 (i-1)^2 \times \frac{P^2(N-P)}{P(N-1)} \sigma_c^2 \\
& + K^2(r-1)^2 P^2 \times \frac{N-P}{P(N-1)} \sigma_c^2 + k^2 \sigma_c^2 \\
& + KR(i-1)K(r-1)P \frac{\sigma_c^2}{N-1} \\
& + 2KR(i-1)kP \times \frac{\sigma_c^2}{N-1} + K(r-1)kP \times \frac{\sigma_c^2}{N-1} \\
& = \left\{ \frac{K^2 R^2 (i-1)^2 P(N-P)}{N-1} + \frac{K^2(r-1)^2(N-P)P}{N-1} \right. \\
& + k^2 - \frac{2K^2 R(i-1)(r-1)P}{N-1} \\
& \left. - \frac{2KR(i-1)kP}{N-1} - \frac{K(r-1)kP}{N-1} \right\} \sigma_c^2
\end{aligned}$$

Then we can bound ⑧:

$$\begin{aligned}
& \sum_{i,j,r,k} \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla F_j(\mathbf{x}^*) \right\|^2 \right] \\
& \leq \frac{K^3 R^3 P^2 \times 2S^3}{6} + \frac{K^3 P^2 S \times 2R^3}{6} + \frac{SPR \times 2K^3}{6} \\
& \leq \frac{1}{3} \underbrace{SPRK^3 (S^2 PR^2 + PR^2 + 1)}_{q_c(S,P,R,K)} \sigma_c^2
\end{aligned}$$

Still follow Lemma 4 in [2], the bound for gradients of edge server ⑨ is given,

$$\begin{aligned}
& \sum_{i,j,r,k} \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\
& \leq \frac{P(P-1)(2P-1)}{6} \times \frac{K(K-1)(2K-1)}{6} \\
& \times \sum_i \sum_r \mathbb{E} \left[\left\| \sum_{i',r'} (\nabla f_i(\mathbf{x}) - \nabla f(\mathbf{x})) \right\|^2 \right]
\end{aligned}$$

$$\begin{aligned}
& \leq \frac{PK(P-1)(K-1)(2P-1)(2K-1)}{36} \\
& \times \frac{1}{2} S^2 R^3 \sigma_g^2 \\
& \leq \underbrace{\frac{1}{18} P^3 K^3 R^3 S^2 \sigma_g^2}_{q_g(S,P,R,K)}
\end{aligned}$$

To bound $\mathbb{E}_t$, we first bound following terms:

$$\begin{aligned}
& \sum_{i,j,r,k} B_{i,j,r,k} \\
& = \sum_{i,j,r,k} \{(i-1)RPK + (r-1)PK + k-1\} \\
& \leq RPK \times \frac{S(S-1)}{2} \times RPK + PK \times SPK \\
& \times \frac{R(R-1)}{2} + SPR + \frac{K(K-1)}{2} \\
& \leq \underbrace{\frac{S(S-1)R^2 P^2 K^2}{2} + \frac{R(R-1)}{2} SP^2 K^2}_{q_B(S,P,R,K)} \\
& + \frac{K(K-1)}{2} SPR
\end{aligned}$$

$$\begin{aligned}
& \sum_{i,j,r,k} B_{i,j,r,k}^2 \\
& \leq \frac{R^3 P^3 K^3 S^2 (S-1)}{K^2 (K-1) SPR} + \frac{SP^3 K^3 \times \frac{R^2 (R-1)}{2}}{R^2 (R-1) S (S-1) P^3 K^3} \\
& + \frac{R^2 P^2 K^2 S (S-1) (K-1)}{2} \\
& + \frac{R(R-1)K^2 (K-1) P^2 S}{2} \\
& \underbrace{\hspace{10em}}_{q_{B^2}(S,P,R,K)}
\end{aligned}$$

$\mathbb{E}_t$ is as,

$$\begin{aligned}
\mathbb{E}_t & \leq 7\eta^2 \sigma^2 \sum_{i,j,r,k} B_{i,j,r,k} \\
& + 7L^2 \eta^2 \sum_{i,j,r,k} B_{i,j,r,k} \sum_{i',j',r',k'} \mathbb{E} \left[\left\| \mathbf{x}_{r',k'}^{i',j'} - \mathbf{x} \right\|^2 \right] \\
& + 14L\eta^2 \sum_{i,j,r,k} B_{i',j',r',k'}^2 D_F(\mathbf{x}, \mathbf{x}^*) \\
& + 7\eta^2 \sum_{i,j,r,k} \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla F_j(\mathbf{x}^*) \right\|^2 \right] \\
& + 42L\eta^2 \sum_{i,j,r,k} B_{i,j,r,k}^2 D_f(\mathbf{x}, \mathbf{x}^*) \\
& + 21\eta^2 \sum_{i,j,r,k} \mathbb{E} \left[\left\| \sum_{i',j',r',k'} \nabla f_i(\mathbf{x}^*) \right\|^2 \right] \\
& \leq 7\eta^2 \sigma^2 q_B + 7L^2 \eta^2 \sum_{i,j,r,k} q_B \mathbb{E}_t \\
& + 14L\eta^2 q_{B^2} D_F(\mathbf{x}, \mathbf{x}^*) \\
& + 7\eta^2 q_c \sigma_c^2 + 42L\eta^2 q_{B^2} D_f(\mathbf{x}, \mathbf{x}^*) + 21\eta^2 q_g \sigma_g^2
\end{aligned}$$

With $7\eta^2\sigma^2q_B \leq 3 \times \frac{7}{2} \times \frac{1}{35^2} = \frac{21}{2450}$, we bound $\mathbb{E}_t$ as,

$$\begin{aligned}\mathbb{E}_t &\leq \frac{2450}{2429} \left\{ 7\eta^2\sigma^2q_B + 14L\eta^2q_{B^2}D_F(\mathbf{x}, \mathbf{x}^*) \right. \\ &\quad \left. + 7\eta^2q\sigma_c^2 + 21\eta^2q_g\sigma_c^2 + 42L\eta^2q_{B^2}D_f(\mathbf{x}, \mathbf{x}^*) \right\} \\ &\leq \frac{71}{10}\eta^2q_B\sigma^2 + \frac{71}{5}Lq_{B^2}\eta^2D_F(\mathbf{x}, \mathbf{x}^*) \\ &\quad + \frac{71}{10}q_c\eta^2\sigma_c^2 + 22q_g\eta^2\sigma_g^2 + 43Lq_{B^2}\eta^2D_f(\mathbf{x}, \mathbf{x}^*)\end{aligned}$$

Theorem 1 For strongly convex objective, under Assumptions 1, 2, 3, we set $\tilde{\eta} := MNRK\eta$, $\Pi_{sc} := \mathbb{E}[f(\mathbf{x}^T) - f(\mathbf{x}^*)]$, $D := \|\mathbf{x}^0 - \mathbf{x}^*\|$, with the following learning rate condition, $\frac{1}{\mu T} \leq \tilde{\eta} \leq \frac{1}{35L}$. We have the upper bound,

$$\begin{aligned}\Pi_{sc} &\leq \underbrace{5\mu D^2 \exp\left(-\frac{\mu\tilde{\eta}T}{2}\right)}_{\text{Optimization term}} + \underbrace{\frac{27\tilde{\eta}\sigma^2}{MNRK} + \frac{53L\sigma_g^2\tilde{\eta}^2}{M}}_{\text{Error terms}} \\ &\quad + \underbrace{\frac{18L\tilde{\eta}^2(M^2NR^2 + NR^2 + 1)\sigma_c^2}{M^2N^2K^2}}_{\text{Error term}}\end{aligned}$$

Proof Based on Lemma 2, substitute $\mathbb{E}_t$ into $\mathbb{E}[\|\mathbf{x}^{t+1} - \mathbf{x}^*\|^2]$,

$$\begin{aligned}\mathbb{E}[\|\mathbf{x}^{t+1} - \mathbf{x}^*\|^2] &\leq \left(1 - \frac{\mu SPKR\eta}{2}\right) \|\mathbf{x} - \mathbf{x}^*\|^2 + 7SPRK\eta^2\sigma^2 \\ &\quad + 7S^2R^2P^2K^2\eta^2 \frac{(N-P)}{P(N-1)}\sigma_c^2 \\ &\quad + \frac{21P^2K^2R^2S^2(M-S)}{S(M-1)}\eta^2\sigma_g^2 - \frac{8}{5}SPKR\eta D_F(\mathbf{x}, \mathbf{x}^*) \\ &\quad + 42LS^2P^2R^2K^2\eta^2 D_f(\mathbf{x}, \mathbf{x}^*) \\ &\quad + \underbrace{(2L\eta + 7L^2SPRK\eta^2)}_{q_{\mathbb{E}_t} \leq \frac{11}{5}L\eta} \underbrace{\sum_{i,j,r,k} \mathbb{E}[\|\mathbf{x}_{r,k}^{i,j} - \mathbf{x}\|^2]}_{\mathbb{E}_t} \\ &\leq \left(1 - \frac{\mu SPKR\eta}{2}\right) \|\mathbf{x} - \mathbf{x}^*\|^2 \\ &\quad + \underbrace{\left[7SPRK\eta^2 + \frac{71}{10}q_Bq_{\mathbb{E}_t}\eta^2\right]}_{\textcircled{1}}\sigma^2 \\ &\quad + \underbrace{\left[7S^2R^2P^2K^2\frac{(N-P)}{P(N-1)}\eta^2 + \frac{71}{10}q_cq_{\mathbb{E}_t}\eta^2\right]}_{\textcircled{2}}\sigma_c^2 \\ &\quad + \underbrace{\left[\frac{21P^2K^2R^2S^2(M-S)}{S(M-1)}\eta^2 + 22q_gq_{\mathbb{E}_t}\eta^2\right]}_{\textcircled{3}}\sigma_g^2 \\ &\quad + \underbrace{\left[q_{D_F} + \frac{71}{5}Lq_{B^2}q_{\mathbb{E}_t}\eta^2\right]}_{\textcircled{4}}D_F(\mathbf{x}, \mathbf{x}^*) \\ &\quad + \underbrace{\left[42LS^2P^2R^2K^2\eta^2 + 43Lq_{B^2}q_{\mathbb{E}_t}\eta^2\right]}_{\textcircled{5}}D_f(\mathbf{x}, \mathbf{x}^*)\end{aligned}$$

Using $\eta \leq \frac{1}{35LSPRK}$ to simplify above equations,

Bounding ①,

$$\begin{aligned}&\left[7SPRK\eta^2 + \frac{71}{10}q_Bq_{\mathbb{E}_t}\eta^2\right]\sigma^2 \\ &\leq \eta^2\sigma^2 \left[7SPRK + \frac{71}{10}q_Bq_{\mathbb{E}_t}\right] \\ &\leq \frac{79}{10}SPRK\eta^2\sigma^2\end{aligned}$$

Bounding ②,

$$\begin{aligned}&\left[7S^2R^2P^2K^2\frac{(N-P)}{P(N-1)}\eta^2 + \frac{71}{10}q_cq_{\mathbb{E}_t}\eta^2\right]\sigma_c^2 \\ &\leq \eta^2\sigma_c^2 \left[\frac{7S^2R^2P^2K^2(N-P)}{P(N-1)} + \frac{71}{10}q_cq_{\mathbb{E}_t}\right] \\ &\leq \left[\frac{7S^2R^2P^2K^2(N-P)}{P(N-1)}\eta^2\right. \\ &\quad \left.+ \frac{53}{10}SPRLK^3(S^2PR^2 + PR^2 + 1)\eta^3\right]\sigma_c^2\end{aligned}$$

Bounding ③,

$$\begin{aligned}&\left[\frac{21P^2K^2R^2S^2(M-S)}{S(M-1)}\eta^2 + 22q_gq_{\mathbb{E}_t}\eta^2\right]\sigma_g^2 \\ &= \frac{21P^2K^2R^2S^2(M-S)}{S(M-1)}\eta^2\sigma_g^2 + 22q_gq_{\mathbb{E}_t}\eta^2\sigma_g^2 \\ &\leq \left[\frac{21P^2K^2R^2S^2(M-S)}{S(M-1)}\eta^2 + 3P^3K^3R^3S^2L\eta^3\right]\sigma_g^2\end{aligned}$$

Bounding ④:

$$\begin{aligned}&\left[q_{D_F} + 22Lq_{B^2}q_{\mathbb{E}_t}\eta^2\right]D_F(\mathbf{x}, \mathbf{x}^*) \\ &\leq \left\{-\frac{8}{5}SPKR\eta + 22 \times \frac{11}{2450}SPRK\eta\right\}D_F(\mathbf{x}, \mathbf{x}^*) \\ &= -\frac{3}{2}SPRK\eta D_F(\mathbf{x}, \mathbf{x}^*)\end{aligned}$$

Bounding ⑤,

$$\begin{aligned}&\left[42LS^2P^2R^2K^2\eta^2 + 42Lq_{B^2}q_{\mathbb{E}_t}\eta^2\right]D_f(\mathbf{x}, \mathbf{x}^*) \\ &\leq 42LS^2P^2R^2K^2\eta^2 + \frac{473}{5}L\eta^2L\eta \\ &\quad \times \left\{\frac{R^2P^3K^3S^2(S-1)}{3} + SP^3K^3 \times \frac{R^2(R-1)}{3} \right. \\ &\quad \left. + \frac{K^2(K-1)SPR}{3} + \frac{R^2(R-1)S(S-1)P^3K^3}{3} \right. \\ &\quad \left. + \frac{R^2P^2K^3S(S-1)(K-1)}{3} + \frac{R(R-1)K^2(K-1)P^2S}{2}\right\} \\ &\leq 45LS^2P^2R^2K^2\eta^2 D_f(\mathbf{x}, \mathbf{x}^*)\end{aligned}$$

In this paper, our algorithm involves one more edge layer to construct HFL, which is the main difference from HL [2], so one more term related to edge server gradient update is gen-

erated. Based on our definition $\nabla f_i(\mathbf{x}) := \frac{1}{P} \sum_{j=0}^{P-1} \nabla F_j(\mathbf{x})$, we have following bound for $D_f(\mathbf{x}, \mathbf{x}^*)$,

$$\begin{aligned} & \mathbb{E} \left[\left\| \sum_{i,j,r,k} [\nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{x}^*)] \right\|^2 \right] \\ &= \mathbb{E} \left[\left\| \sum_{i,j,r,k} \left[\frac{1}{P} \sum_{j=0}^{P-1} \nabla F_j(\mathbf{x}) - \frac{1}{P} \sum_{j=0}^{P-1} \nabla F_j(\mathbf{x}^*) \right] \right\|^2 \right] \\ &= \mathbb{E} \left[\left\| \frac{1}{P} \sum_{j=0}^{P-1} \sum_{i,j,r,k} [\nabla F_j(\mathbf{x}) - \nabla F_j(\mathbf{x}^*)] \right\|^2 \right] \\ &\leq \mathbb{E} \left[\left\| \sum_{i,j,r,k} [\nabla F_j(\mathbf{x}) - \nabla F_j(\mathbf{x}^*)] \right\|^2 \right] \end{aligned}$$

Then we have following bound,

$$\begin{aligned} & \mathbb{E} [\|\mathbf{x}^{t+1} - \mathbf{x}^*\|^2] \\ &\leq \left(1 - \frac{\mu SPKR\eta}{2}\right) \|\mathbf{x}^t - \mathbf{x}^*\|^2 + \frac{71}{10} SPKR\eta^2\sigma^2 \\ &\quad + \left[\frac{7S^2R^2P^2K^2(N-P)}{P(N-1)}\eta^2 \right. \\ &\quad + \left. \frac{53}{10} SPRLK^3(S^2PR^2 + PR^2 + 1)\eta^3 \right] \sigma_c^2 \\ &\quad + \left[\frac{21P^2K^2R^2S^2(M-S)}{S(M-1)}\eta^2 + 3P^3K^3R^3S^2L\eta^3 \right] \sigma_g^2 \\ &\quad - \frac{3}{2} SPKR\eta D_F(\mathbf{x}, \mathbf{x}^*) \\ &\quad + 45LS^2P^2R^2K^2\eta^2 D_f(\mathbf{x}, \mathbf{x}^*) \end{aligned}$$

Let $\tilde{\eta} = SPKR\eta$,

$$\begin{aligned} & \mathbb{E} [\|\mathbf{x}^{t+1} - \mathbf{x}^*\|^2] \\ &\leq \left(1 - \frac{\mu\tilde{\eta}}{2}\right) \|\mathbf{x}^t - \mathbf{x}^*\|^2 + \frac{79}{10} \frac{\tilde{\eta}^2}{SPKR} \sigma^2 \\ &\quad + \left[\frac{7\eta^2(N-P)}{P(N-1)} + \frac{53}{10} \tilde{\eta}^3 L \left(\frac{1}{P} + \frac{1}{S^2P} + \frac{1}{S^3P^2R^2} \right) \right] \sigma_c^2 \\ &\quad + \left[\frac{21\tilde{\eta}^2(M-S)}{S(M-1)} + \frac{3L\tilde{\eta}^3}{S} \right] \sigma_g^2 - \frac{3}{10} \tilde{\eta} D_F(\mathbf{x}, \mathbf{x}^*) \end{aligned} \quad (1)$$

Based on Lemma 7 in [2], with $a = \frac{\mu}{2}$, $b = \frac{3}{10}$, $S_t = D_F(\mathbf{x}, \mathbf{x}^*)$, $C_1 = \frac{79}{10} \times \frac{\sigma^2}{SPKR} + \frac{7(N-P)}{P(N-1)}\sigma_c^2 + \frac{21(M-S)}{S(M-1)}\sigma_g^2$, $C_2 = \frac{53}{10} \left(\frac{1}{P} + \frac{1}{S^2P} + \frac{1}{S^3P^2R^2} \right) L\sigma_c^2 + \frac{3L}{S}\sigma_g^2$, with $w_t = \left(1 - \frac{\mu\tilde{\eta}}{2}\right)^{-(t+1)}$ we have,

$$\begin{aligned} & \mathbb{E} [\|\mathbf{x}^{t+1} - \mathbf{x}^*\|^2] \\ &\leq 5\mu \|\mathbf{x}^0 - \mathbf{x}^*\|^2 \exp\left(-\frac{\mu}{2}\tilde{\eta}T\right) \\ &\quad + \left\{ \frac{27\sigma^2}{SPKR} + \frac{24(N-P)}{P(N-1)}\sigma_c^2 + \frac{70(M-S)}{S(M-1)}\sigma_g^2 \right\} \tilde{\eta} \\ &\quad + \left\{ 18L \left(\frac{1}{P} + \frac{1}{S^2P} + \frac{1}{S^3P^2R^2} \right) \sigma_c^2 + \frac{53L}{S}\sigma_g^2 \right\} \times \tilde{\eta}^2 \end{aligned}$$

When $S = M, P = N$ for edge and client fully participation with $D := \|\mathbf{x}^0 - \mathbf{x}^*\|$, we get,

$$\begin{aligned} & \mathbb{E} [f(\mathbf{x}^T) - f(\mathbf{x}^*)] \\ &\leq 5\mu D^2 \exp\left(-\frac{\mu\tilde{\eta}T}{2}\right) + \frac{27\tilde{\eta}\sigma^2}{MNRK} \\ &\quad + \frac{18L\tilde{\eta}^2(M^2NR^2 + NR^2 + 1)\sigma_c^2}{M^2N^2K^2} + \frac{53L\sigma_g^2\tilde{\eta}^2}{M} \end{aligned}$$

By turning the learning rate carefully, we get:

$$\begin{aligned} & \mathbb{E} [f(\mathbf{x}^T) - f(\mathbf{x}^*)] \\ &= \tilde{O} \left(\mu D \exp\left(-\frac{\mu T}{70L}\right) + \frac{\sigma^2}{SPKR\mu T} \frac{(N-P)\sigma_c^2}{\mu TP(N-1)} \right. \\ &\quad + \frac{(M-S)}{S(M-1)\mu T} \sigma_g^2 + \frac{L\sigma_g^2}{S\mu^2 T^2} \\ &\quad \left. + \frac{\left(\frac{1}{P} + \frac{1}{S^2P} + \frac{1}{S^3P^2R^2}\right) L\sigma_c^2}{\mu^2 T^2} \right) \end{aligned}$$

When $P = N, S = M$ with edge and client fully participation, we have,

$$\begin{aligned} & \mathbb{E} [f(\mathbf{x}^T) - f(\mathbf{x}^*)] \\ &= \tilde{O} \left(\frac{\sigma^2}{\mu MNRKT} + \frac{L(M^2NR^2 + NR^2 + 1)\sigma_c^2}{M^2N^2K^2\mu^2 T^2} \right. \\ &\quad \left. + \frac{L\sigma_g^2}{M\mu^2 T^2} + \mu D^2 \exp\left(-\frac{\mu T}{70L}\right) \right) \end{aligned}$$

Proofs of Theorem 2 for General Convex Objective

Theorem 2 For general convex objective, under Assumptions 1, 2, 3, set $\tilde{\eta} := MNRK\eta$, $\Pi_{gc} := \mathbb{E} [f(\mathbf{x}^T) - f(\mathbf{x}^*)]$, $D := \|\mathbf{x}^0 - \mathbf{x}^*\|$, With the following learning rate condition, $\tilde{\eta} \leq \frac{1}{35L}$. We have the upper bound,

$$\begin{aligned} \Pi_{gc} &\leq \underbrace{\frac{10D^2}{3\tilde{\eta}T}}_{\text{Optimization term}} + \underbrace{\frac{27\tilde{\eta}\sigma^2}{MNRK} + \frac{53L\sigma_g^2\tilde{\eta}^2}{M}}_{\text{Error terms}} \\ &\quad + \underbrace{\frac{18L\tilde{\eta}^2(M^2NR^2 + NR^2 + 1)\sigma_c^2}{M^2N^2K^2}}_{\text{Error term}} \end{aligned}$$

Proof For general case: When $\mu = 0$, the following bound based on Eq.1 can be given,

$$\begin{aligned} & \mathbb{E} [\|\mathbf{x}^{t+1} - \mathbf{x}^*\|^2] \\ &\leq \|\mathbf{x}^t - \mathbf{x}^*\|^2 + \frac{79}{10} \frac{\tilde{\eta}^2}{SPKR} \sigma^2 \\ &\quad + \left[\frac{7\eta^2(N-P)}{P(N-1)} + \frac{53}{10} \tilde{\eta}^3 L \left(\frac{1}{P} + \frac{1}{S^2P} + \frac{1}{S^3P^2R^2} \right) \right] \sigma_c^2 \\ &\quad + \left[\frac{21\tilde{\eta}^2(M-S)}{S(M-1)} + \frac{3L\tilde{\eta}^3}{S} \right] \sigma_g^2 - \frac{3}{10} \tilde{\eta} D_F(\mathbf{x}, \mathbf{x}^*) \end{aligned}$$

Applying Lemma 8 in [2] with $b = \frac{3}{10}$, $\frac{1}{d} = \frac{1}{35L}$, $\frac{\gamma_0}{\gamma Tb} = \frac{10}{\tilde{\eta}T^3} \|\mathbf{x}^0 - \mathbf{x}^*\|^2$, $C_1\gamma = \frac{C_1\tilde{\eta}}{b} =$

$\left\{ \frac{27\sigma^2}{SPKR} + \frac{24(P-P)}{P(N-1)}\sigma_c^2 + \frac{70(M-S)}{S(M-1)}\sigma_g^2 \right\} \tilde{\eta}, C_2\gamma^2 = \frac{C_2\tilde{\eta}}{b} =$ When $P = N, S = M$ with edge and client fully participation, we have,

$\left\{ 18L \left(\frac{1}{P} + \frac{1}{S^2P} + \frac{1}{S^2P^2R^2} \right) \sigma_c^2 + \frac{53L}{S}\sigma_g^2 \right\} \times \tilde{\eta}^2$, we get,

$$\begin{aligned} & \mathbb{E} \left[\left\| \mathbf{x}^{t+1} - \mathbf{x}^* \right\|^2 \right] \\ & \leq \frac{10}{\tilde{\eta}T^3} \left\| \mathbf{x}^0 - \mathbf{x}^* \right\|^2 \\ & \quad + \left\{ \frac{27\sigma^2}{SPKR} + \frac{24(N-P)}{P(N-1)}\sigma_c^2 + \frac{70(M-S)}{S(M-1)}\sigma_g^2 \right\} \tilde{\eta} \\ & \quad + \left\{ 18L \left(\frac{1}{P} + \frac{1}{S^2P} + \frac{1}{S^2P^2R^2} \right) \sigma_c^2 + \frac{53L}{S}\sigma_g^2 \right\} \times \tilde{\eta}^2 \end{aligned}$$

When $P = N, S = M$ with edge and client fully participation with $D := \left\| \mathbf{x}^0 - \mathbf{x}^* \right\|$, we have,

$$\begin{aligned} & \mathbb{E} [f(\mathbf{x}^T) - f(\mathbf{x}^*)] \\ & \leq \frac{10D^2}{3\tilde{\eta}T} + \frac{27\tilde{\eta}\sigma^2}{MNRK} + \frac{18L\tilde{\eta}^2(M^2NR^2 + NR^2 + 1)\sigma_c^2}{M^2N^2K^2} \\ & \quad + \frac{53L\sigma_g^2\tilde{\eta}^2}{M} \end{aligned}$$

Then tuning learning rate carefully,

$$\begin{aligned} & 2C_1^{\frac{1}{2}} \left(\frac{\gamma_0}{T} \right)^{\frac{1}{2}} \\ & = 2 \left\{ \frac{\sqrt{27}\sigma}{\sqrt{SPKR}} + \sqrt{\frac{RP}{P(N-1)}}\sigma_c + \sqrt{\frac{M-S}{(M-1)S}}\sigma_g \right\} \times \frac{D}{\sqrt{T}} \\ & = O \left(\frac{\sigma D}{\sqrt{SPRKT}} + \sqrt{\frac{N-P}{P(N-1)T}}D\sigma_c + \sqrt{\frac{M-S}{(M-1)ST}}D\sigma_g \right) \\ & 2C_2^{\frac{1}{3}} \left(\frac{\gamma_0}{T} \right)^{\frac{2}{3}} \\ & = 2 \left\{ 18L \left(\frac{1}{P} + \frac{1}{S^2P} + \frac{1}{S^2P^2R^2} \right) \sigma_c^2 + \frac{53L}{M}\sigma_g^2 \right\}^{\frac{1}{3}} \frac{D^{\frac{4}{3}}}{T^{\frac{2}{3}}} \\ & = O \left(\frac{(LD^4\sigma_c^2)^{\frac{1}{3}}}{P^{\frac{1}{3}}T^{\frac{2}{3}}} + \frac{(L\sigma_c^2D^4)^{\frac{1}{3}}}{(ST)^{\frac{2}{3}}P^{\frac{1}{3}}} + \frac{(L\sigma_c^2D^4)^{\frac{1}{3}}}{(SPRT)^{\frac{2}{3}}} + \frac{(L\sigma_g^2D^4)^{\frac{1}{3}}}{M^{\frac{1}{3}}T^{\frac{2}{3}}} \right) \end{aligned}$$

$$\frac{d\gamma_0}{T+1} = \frac{35LD^2}{T} = O \left(\frac{LD^2}{T} \right)$$

Then,

$$\begin{aligned} & \mathbb{E} [f(\mathbf{x}^T) - f(\mathbf{x}^*)] \\ & = O \left(\frac{\sigma D}{\sqrt{SPRKT}} + \sqrt{\frac{N-P}{P(N-1)T}}D\sigma_c + \sqrt{\frac{M-S}{(M-1)ST}}\sigma_g \right. \\ & \quad + \frac{(LD^4\sigma_c^2)^{\frac{1}{3}}}{P^{\frac{1}{3}}T^{\frac{2}{3}}} + \frac{(L\sigma_c^2D^4)^{\frac{1}{3}}}{(ST)^{\frac{2}{3}}P^{\frac{1}{3}}} + \frac{(L\sigma_c^2D^4)^{\frac{1}{3}}}{(SPRT)^{\frac{2}{3}}} \\ & \quad \left. + \frac{(L\sigma_g^2D^4)^{\frac{1}{3}}}{M^{\frac{1}{3}}T^{\frac{2}{3}}} + \frac{LD^2}{T} \right) \end{aligned}$$

$$\begin{aligned} & \mathbb{E} [f(\mathbf{x}^{(T)}) - f(\mathbf{x}^*)] \\ & = O \left(\frac{\sigma D}{\sqrt{MNRKT}} + \frac{(L(M^2NR^2 + NR^2 + 1)D^4\sigma_c^2)^{\frac{1}{3}}}{(MNR)^{2/3}} \right. \\ & \quad \left. + \frac{(L\sigma_g^2D^4)^{\frac{1}{3}}}{(MT^2)^{\frac{1}{3}}} + \frac{LD^2}{T} \right) \end{aligned}$$

Proofs of Theorem 3 for Non-Convex Objective

Lemma 3 With Assumption 1,2,4, the learning rate satisfies $\eta \leq \frac{1}{35LSKR\bar{P}}$, the global model updates after one global round should be bounded as follows:

$$\begin{aligned} \mathbb{E} [F(\mathbf{x}^{t+1}) - F(\mathbf{x}^t)] & \leq -\frac{SRKP\eta}{2} \mathbb{E} [\left\| \nabla F(\mathbf{x}^t) \right\|^2] \\ & \quad + L\eta^2 SRKP\sigma^2 \\ & \quad + \frac{L^2\eta}{2} \sum_{i,j,r,k} \mathbb{E} [\left\| \mathbf{x}_{r,k}^{i,j,t} - \mathbf{x}^t \right\|^2] \end{aligned}$$

Proof For CHFL, the model updates of one global round is,

$$\begin{aligned} \Delta \mathbf{x} = \mathbf{x}^{t+1} - \mathbf{x}^t & = -\eta \sum_{i=1}^S \sum_{j=1}^P \sum_{r=0}^{R-1} \sum_{k=0}^{K-1} g_{r,k}^{i,j} \\ & = -\eta \sum_{i,j,r,k} g_{r,k}^{i,j} \end{aligned}$$

where $\mathbf{g}_{r,k}^{i,j} = \nabla F_j(\mathbf{x}_{r,k}^{i,j}, \xi_{r,k}^{i,j})$, thus,

$$\mathbb{E}[\Delta \mathbf{x}] = -\eta \sum_{i,j,r,k} \mathbb{E} [\nabla F_j(\mathbf{x}_{r,k}^{i,j})]$$

We focus on a single training round and drop the superscripts t :

$$\begin{aligned} & \mathbb{E}[F(\mathbf{x} + \Delta \mathbf{x}) - F(\mathbf{x})] \\ & \leq \mathbb{E}[\langle \nabla F(\mathbf{x}), \Delta \mathbf{x} \rangle] + \frac{L}{2} \mathbb{E} [\left\| \Delta \mathbf{x} \right\|^2] \\ & \leq \underbrace{-\eta \sum_{i,j,r,k} \mathbb{E} [\langle \nabla F(\mathbf{x}), \nabla F_j(\mathbf{x}_{r,k}^{i,j}) \rangle]}_A \\ & \quad + \underbrace{\frac{L\eta^2}{2} \mathbb{E} \left[\left\| \sum_{i,j,r,k} g_{r,k}^{i,j} \right\|^2 \right]}_B \end{aligned}$$

Bounding A,

$$\begin{aligned}
& -\eta \sum_{i,j,r,k} \mathbb{E} \left[\left\langle \nabla F(\mathbf{x}), \nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) \right\rangle \right] \\
& = -\frac{\eta}{2} \sum_{i,j,r,k} \mathbb{E} \left[\|\nabla F(\mathbf{x})\|^2 + \|\nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right)\|^2 \right. \\
& \quad \left. - \|\nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) - \nabla F(\mathbf{x})\|^2 \right] \\
& \leq -\frac{\eta SPRK}{2} \|\nabla F(\mathbf{x})\|^2 - \frac{\eta}{2} \sum_{i,j,r,k} \mathbb{E} \|\nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right)\|^2 \\
& \quad + \frac{L^2 \eta}{2} \sum_{i,j,r,k} \mathbb{E} \left[\|\mathbf{x}_{r,k}^{i,j} - \mathbf{x}\|^2 \right]
\end{aligned}$$

Bounding B,

$$\begin{aligned}
& \frac{L\eta^2}{2} \mathbb{E} \left[\left\| \sum_{i,j,r,k} g_{r,k}^{i,j} \right\|^2 \right] \\
& \leq \frac{L^2 \eta}{2} \mathbb{E} \left[\left\| \sum_{i,j,r,k} g_{r,k}^{i,j} - \sum_{i,j,r,k} \nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) + \sum_{i,j,r,k} \nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right) \right\|^2 \right] \\
& \leq L\eta^2 SPRK \sigma^2 + LSPRK \eta^2 \sum_{i,j,r,k} \mathbb{E} \left[\|\nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right)\|^2 \right]
\end{aligned}$$

When $\eta \leq \frac{1}{35LSPKR}$, we have $-\frac{\eta}{2}(1 - 2L\eta SKRP) \sum_{i,j,r,k} \mathbb{E} \left[\|\nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right)\|^2 \right] < 0$, thus,

$$\begin{aligned}
& \mathbb{E}[F(\mathbf{x} + \Delta \mathbf{x}) - F(\mathbf{x})] \\
& \leq -\frac{\eta SKRP}{2} \|\nabla F(\mathbf{x})\|^2 + L\eta^2 SRKP \sigma^2 \\
& \quad + \frac{L^2 \eta}{2} \sum_{i,r,k,j} \mathbb{E} \left[\|\mathbf{x}_{r,k}^{i,j} - \mathbf{x}\|^2 \right] \\
& \quad - \frac{\eta}{2} \sum_{i,r,k,j} \mathbb{E} \left[\|\nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right)\|^2 \right] \\
& \quad + L\eta^2 SRKP \sum_{i,r,k,j} \mathbb{E} \left[\|\nabla F_j \left(\mathbf{x}_{r,k}^{i,j} \right)\|^2 \right] \\
& \leq -\frac{SRKP}{2} \|\nabla F(\mathbf{x})\|^2 + L\eta^2 SRKP \sigma^2 \\
& \quad + \underbrace{\frac{L^2 \eta}{2} \sum_{i,r,k,j} \mathbb{E} \left[\|\mathbf{x}_{r,k}^{i,j} - \mathbf{x}\|^2 \right]}_{\text{client drift}}
\end{aligned}$$

Lemma 4 Let Assumptions 1,2,4 hold and assume that all the local objectives are μ -strongly convex. If the learning rate satisfies $\eta \leq \frac{1}{35LSPRTK}$, then the client shift can be bounded as:

$$\begin{aligned}
\mathbb{E}_t \leq & 6\eta^2 q_B \sigma^2 + 6\eta^2 q_{B^2} \sigma_c^2 + 6\eta^2 q_{B^2} \sigma_g^2 \\
& + 6\eta^2 q_{B^2} \|\nabla f(\mathbf{x})\|^2
\end{aligned}$$

where q_B and q_{B^2} can be found in Lemma 2.

Proof To bound $\mathbb{E}_t$, we first bound $\mathbb{E} \left[\|\mathbf{x}_{r,k}^{i,j} - \mathbf{x}\|^2 \right]$,

$$\begin{aligned}
& \mathbb{E} \left[\|\mathbf{x}_{r,k}^{i,j} - \mathbf{x}\|^2 \right] \\
& = \eta^2 \mathbb{E} \left[\left\| \underbrace{\sum_{i'=1}^i \sum_{r'=0}^{r(i)} \sum_{j'=0}^{j-1} \sum_{k'=0}^{k(i)} g_{r',k'}^{i',j'}}_{B_{i,r,j,k}} \right\|^2 \right] \\
& \leq \eta^2 \mathbb{E} \left[\left\| B_{i,j,r,k} \left[g_{r',k'}^{i',j'} - \nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) \right. \right. \right. \\
& \quad \left. \left. + \nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) - \nabla F_j(\mathbf{x}) + \nabla F_j(\mathbf{x}) - \nabla f'_i(\mathbf{x}) \right. \right. \\
& \quad \left. \left. + \nabla f'_i(\mathbf{x}) - \nabla f(\mathbf{x}) + \nabla f(\mathbf{x}) \right] \right\|^2 \right] \\
& \leq 5\eta^2 B_{i,j,r,k} \mathbb{E} \left[\left\| g_{r',k'}^{i',j'} - \nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) \right\|^2 \right] \\
& \quad + 5\eta^2 B_{i,j,r,k} \sum_{i',r',j',k'} \mathbb{E} \left[\left\| \nabla F_j \left(\mathbf{x}_{r',k'}^{i',j'} \right) - \nabla F_j(\mathbf{x}) \right\|^2 \right] \\
& \quad + 5\eta^2 B_{i,j,r,k} \sum_{i',r',j',k'} \mathbb{E} \left[\left\| \nabla F_j(\mathbf{x}) - \nabla f'_i(\mathbf{x}) \right\|^2 \right] \\
& \quad + 5\eta^2 B_{i,j,r,k} \sum_{i',r',j',k'} \mathbb{E} \left[\left\| \nabla f'_i(\mathbf{x}) - \nabla f(\mathbf{x}) \right\|^2 \right] \\
& \quad + 5\eta^2 B_{i,j,r,k}^2 \|\nabla f(\mathbf{x})\|^2
\end{aligned}$$

Bounding $\mathbb{E}_t$,

$$\begin{aligned}
\mathbb{E}_t \leq & 5\eta^2 \sigma^2 \sum_{i,j,r,k} B_{i,j,r,k} \\
& + 5\eta^2 L^2 \sum_{i,j,r,k} B_{i,j,r,k}^2 \mathbb{E} \left[\|\mathbf{x}_{r',k'}^{i',j'} - \mathbf{x}\|^2 \right] \\
& + 5\eta^2 \sigma_c^2 \sum_{i,j,r,k} B_{i,j,r,k}^2 + 5\eta^2 \sigma_g^2 \sum_{i,j,r,k} B_{i,j,r,k}^2 \\
& + 5\eta^2 \sum_{i,j,r,k} B_{i,j,r,k}^2 \|\nabla f(\mathbf{x})\|^2 \\
& \leq 5\eta^2 \sigma^2 q_B + 5\eta^2 L^2 q_B \mathbb{E}_t + 5\eta^2 \sigma_c^2 q_{B^2} \\
& \quad + 5\eta^2 \sigma_g^2 q_{B^2} + 5\eta^2 q_{B^2} \|\nabla f(\mathbf{x})\|^2
\end{aligned}$$

$$(1 - 5\eta^2 L^2 q_B) \mathbb{E}_t \leq 5\eta^2 \sigma^2 q_B + 5\eta^2 \sigma_c^2 q_{B^2} + 5\eta^2 \sigma_g^2 q_{B^2} + 5\eta^2 q_{B^2} \|\nabla f(\mathbf{x})\|^2$$

With the condition $\eta \leq \frac{1}{35SPRKL}$, we have $5L^2 q_B \eta^2 \leq \frac{1}{490}$, thus,

$$\begin{aligned}
\mathbb{E}_t \leq & \frac{490}{489} \times \{5\eta^2 \sigma^2 A + 5\eta^2 \sigma_c^2 B \\
& + 5\eta^2 \sigma_g^2 B + 5\eta^2 B \|\nabla f(\mathbf{x})\|^2\} \\
& = 6\eta^2 q_B \sigma^2 + 6\eta^2 \sigma_c^2 q_{B^2} + 6\eta^2 q_{B^2} \sigma_g^2 \\
& \quad + 6\eta^2 q_{B^2} \|\nabla f(\mathbf{x})\|^2
\end{aligned}$$

Theorem 3 For CHFL with non-convex objective, under Assumptions 1, 2, and 4, $\Pi_{nc} := \min_{0 \leq t \leq T} \mathbb{E} \left[\|\nabla f(\mathbf{x}^{(t)})\|^2 \right]$, we have the following upper bounds, With the following learning rate condition, $\tilde{\eta} \leq \frac{1}{35L}$. We have the upper bound,

$$\Pi_{nc} \leq \underbrace{\frac{2[f(\mathbf{x}^0) - f(\mathbf{x}^*)]}{\tilde{\eta}T}}_{\text{Optimization term}} + \underbrace{\frac{12L\tilde{\eta}\sigma^2}{5MNRK}}_{\text{Error term}} + \underbrace{\frac{2L^2q_\sigma(M, N, R, K)}{MNRK}(\sigma_g^2 + \sigma_c^2)\tilde{\eta}^2}_{\text{Error terms}} \quad (2)$$

where $q_\sigma(M, N, R, K) = \frac{RNK(M-1)}{2R+3(R-1)(M-1)NK+3(M-1)(K-1)+\frac{2(K-1)}{RN}+3(K-1)} + \frac{3}{6M}$.

Based on Lemma 4, substitute $\mathbb{E}_t$ into $\mathbb{E}[F(\mathbf{x}^{t+1}) - F(\mathbf{x}^t)]$,

$$\begin{aligned} & \mathbb{E}[F(\mathbf{x}^{t+1}) - F(\mathbf{x}^t)] \\ & \leq -\frac{SRKP\eta}{2}\|\nabla F(\mathbf{x}^t)\|^2 + L\eta^2SRKP\sigma^2 \\ & \quad + \underbrace{\frac{L^2\eta}{2}\sum_{i,r,k,j}\mathbb{E}\left[\|\mathbf{x}_{r,k}^{i,j} - \mathbf{x}\|^2\right]}_{\mathbb{E}_t} \\ & \leq -\frac{SRKP\eta}{2}\|\nabla F(\mathbf{x}^t)\|^2 \\ & \quad + \frac{L^2\eta}{2} \times 6\eta^2q_{B^2}\|\nabla f(\mathbf{x}^t)\|^2 \\ & \quad + L\eta^2SRKP\sigma^2 + \frac{L^2\eta}{2} \times 6\eta^2q_B\sigma_g^2 + \frac{L^2\eta}{2} \\ & \quad \times 6\eta^2q_{B^2}\sigma_c^2 + \frac{L^2\eta}{2} \times 6\eta^2q_{B^2}\sigma_g^2 \\ & \leq -\frac{1}{2}SPRK\eta\|\nabla F(\mathbf{x}^t)\|^2 + (LSRP\eta^2 \\ & \quad + 3L^2\eta^3q_B)\sigma^2 + 3L^2\eta^3q_{B^2}\sigma_c^2 + 3L^2\eta^3q_{B^2}\sigma_g^2 \end{aligned}$$

Subtracting F^* for both side,

$$\begin{aligned} \mathbb{E}[F(\mathbf{x}^{t+1}) - F^*] & \leq \mathbb{E}[F(\mathbf{x}^t) - F^*] - \frac{1}{2}\tilde{\eta}\mathbb{E}[\|\nabla F(\mathbf{x}^t)\|^2] \\ & \quad + \underbrace{(LSRP\eta^2 + 3L^2\eta^3q_B)\sigma^2}_{\textcircled{1}} \\ & \quad + \underbrace{3L^2\eta^3q_{B^2}(\sigma_c^2 + \sigma_g^2)}_{\textcircled{2}} \end{aligned}$$

With $\eta \leq \frac{1}{35SPRKL}$, we simplify $\textcircled{1}$,

$$\begin{aligned} & (LSRP\eta^2 + 3L^2\eta^3q_B)\sigma^2 \\ & = LSRKP\eta^2\sigma^2 \left[1 + 3L\eta \left[\frac{(S-1)RPK}{2} \right. \right. \\ & \quad \left. \left. + \frac{(R-1)PK}{2} + \frac{(K-1)}{2} \right] \right] \\ & \leq LSRKP\eta^2\sigma^2 \left[1 + 3L \times \frac{1}{35SPRKL} \right. \\ & \quad \left. \left[\frac{(S-1)RPK}{2} + \frac{(R-1)PK}{2} + \frac{(K-1)}{2} \right] \right] \\ & \leq LSRKP\eta^2\sigma^2 \left[1 + 3 \times \frac{1}{35} \left[\frac{1}{2} + \frac{1}{2} + \frac{1}{2} \right] \right] \\ & \leq \frac{6}{5}LSRP\eta^2\sigma^2 \end{aligned}$$

Using $\tilde{\eta} = SRKP\eta$, then $(LSRP\eta^2 + 3L^2\eta^3q_B)\sigma^2 \leq \frac{6}{5}\frac{L\tilde{\eta}^2\sigma^2}{SRKP}$. With $\eta \leq \frac{1}{35SPRKL}$, we simplify $\textcircled{2}$,

$$\begin{aligned} & 3L^2\eta^3q_{B^2}(\sigma_c^2 + \sigma_g^2) \\ & = \frac{3L^2q_{B^2}}{S^3P^3K^3R^3}(\sigma_c^2 + \sigma_g^2)\eta^3 \\ & \leq 3L^2 \times \left\{ \begin{aligned} & \frac{R^3P^3K^3S^2(S-1)}{3} + \frac{SP^3K^3R^2(R-1)}{3} \\ & + \frac{K^2(K-1)SPR}{2} + \frac{R^2(R-1)S(S-1)P^3K^3}{2} \\ & + \frac{R^2P^2K^2S(S-1)(K-1)}{2} + \frac{R(R-1)K^2(K-1)P^2S}{2} \end{aligned} \right\} \\ & \quad \times \frac{(\sigma_c^2 + \sigma_g^2)}{S^3P^3K^3R^3} \\ & \leq 3L^2 \times \left\{ \frac{RPK(S-1)}{3} + \frac{R-1}{S^3} + \frac{K-1}{3SPR} \right. \\ & \quad + \frac{(R-1)(S-1)PK}{2S} + \frac{(S-1)(K-1)}{2S} \\ & \quad \left. + \frac{(R-1)(K-1)}{2SR} \right\} \times \frac{1}{SPKR}q_\sigma(SP, RK) \\ & = \frac{3L^2q_\sigma(S, P, R, K)(\sigma_g^2 + \sigma_c^2)}{SPRK}\tilde{\eta}^3 \end{aligned}$$

Follow Lemma 8 in [2], we have $r_0 = F(\mathbf{x}^0) - F(\mathbf{x}^*)$, $\gamma = \tilde{\eta}$, $b = \frac{1}{2}$, $C_1 = \frac{6}{5}\frac{L\sigma^2}{SRKP}$, $C_2 = \frac{3L^2q_\sigma(S, P, R, K)(\sigma_g^2 + \sigma_c^2)}{SPRK}$, we have,

$$\begin{aligned} & \min_{0 \leq t \leq T} [\|\nabla F(\mathbf{x}^t)\|^2] \\ & \leq \frac{r_0}{b\gamma(T+1)} + \frac{C_1\gamma}{b} + \frac{C_2\gamma^2}{b} \\ & \leq \frac{2}{\tilde{\eta}T} [F(\mathbf{x}^0) - F(\mathbf{x}^*)] + 2C_1\tilde{\eta} + 2C_2\tilde{\eta}^2 \\ & \leq \frac{2}{\tilde{\eta}T} [F(\mathbf{x}^0) - F(\mathbf{x}^*)] + \frac{12\sigma^2}{5SRKP}\tilde{\eta} + \frac{6L^2q_\sigma(S, P, R, K)}{SPRK}\tilde{\eta}^2 \end{aligned}$$

The convergence rate of non-convex case is followed and $H := f(\mathbf{x}^0) - f(\mathbf{x}^*)$,

$$\begin{aligned} & \min_{0 \leq t \leq T} \mathbb{E}[\|\nabla F(\mathbf{x}^t)\|^2] \\ & = \mathcal{O} \left(\frac{(L\sigma^2H)^{\frac{1}{2}}}{\sqrt{SPRKT}} + \frac{(L^2H^2q_\sigma(\sigma_g^2 + \sigma_c^2))^{\frac{1}{3}}}{(SPRKT^2)^{\frac{1}{3}}} + \frac{LH}{T} \right) \end{aligned}$$

When $P = N, S = M$ with edge and client fully participation, we have,

$$\begin{aligned} & \min_{0 \leq t \leq T} \mathbb{E}[\|\nabla f(\mathbf{x}^t)\|^2] \\ & = \mathcal{O} \left(\frac{(L\sigma^2H)^{1/2}}{\sqrt{MNRKT}} + \frac{(L^2q_\sigma H^2)^{1/3}}{(MNRKT^2)^{1/3}}(\sigma_g^2 + \sigma_c^2)^{\frac{1}{3}} + \frac{LH}{T} \right) \end{aligned}$$

REFERENCES

- [1] Y. J. Cho, P. Sharma, G. Joshi, Z. Xu, S. Kale, and T. Zhang, "On the convergence of federated averaging with cyclic client participation," *arXiv preprint arXiv:2302.03109*, 2023.
- [2] Y. Li and X. Lyu, "Convergence analysis of sequential federated learning on heterogeneous data," *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [3] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh, "Scaffold: Stochastic controlled averaging for federated learning," in *International conference on machine learning*. PMLR, 2020, pp. 5132–5143.